

Evaluations by peers and by outside evaluators are common enough that we have formalized processes to guide the evaluation. These processes define who should participate and which activities should occur during the evaluation. Formalizing a process enables the organization to make the process more repeatable, help the stakeholders understand what will be required and delivered by the evaluation, train new evaluators to use the process, and understand the investment required to perform the evaluation.

We begin by describing a process for outside evaluators (Architecture Tradeoff Analysis Method); we then describe a process for peer review (Lightweight Architecture Evaluation).

21.5 The Architecture Tradeoff Analysis Method

The Architecture Tradeoff Analysis Method (ATAM) is the process we have formalized to perform architecture evaluations. The ATAM has been used for more than two decades to evaluate software architectures of large systems in domains ranging from automotive to financial to defense. The ATAM is designed so that evaluators do not need prior familiarity with the architecture or its business goals, and the system need not be constructed yet. An ATAM exercise may be held either in person or remotely.

Participants in the ATAM

The ATAM requires the participation and mutual cooperation of three groups:

- *The evaluation team.* This group is external to the project whose architecture is being evaluated. It usually consists of three to five people. Each member of the team is assigned a number of specific roles to play during the evaluation; a single person may adopt several roles in an ATAM exercise. (See Table 21.1 for a description of these roles.) The evaluation team may be a standing unit in which architecture evaluations are regularly performed, or its members may be chosen from a pool of architecturally savvy individuals for the occasion. They may work for the same organization as the development team whose architecture is on the table, or they may be outside consultants. In any case, they need to be recognized as competent, unbiased outsiders with no hidden agendas or axes to grind.
- *Project decision makers.* These people are empowered to speak for the development project or have the authority to mandate changes to it. They usually include the project manager and, if an identifiable customer is footing the bill for the development, a representative of that customer may be present as well. The architect is always included—a cardinal rule of architecture evaluation is that the architect must willingly participate.
- *Architecture stakeholders.* Stakeholders have a vested interest in the architecture performing as advertised. They are the people whose ability to do their job hinges on the architecture promoting modifiability, security, high reliability, or the like. Stakeholders include developers, testers, integrators, maintainers, performance engineers, users, and

builders of systems interacting with the one under consideration. Their job during an evaluation is to articulate the specific quality attribute goals that the architecture should meet for the system to be considered a success. A rule of thumb—and that is all it is—is that you should expect to enlist 10 to 25 stakeholders for the evaluation of a large enterprise-critical architecture. Unlike the evaluation team and the project decision makers, stakeholders do not participate in the entire exercise.

TABLE 21.1 ATAM Evaluation Team Roles

Role	Responsibilities
Team Leader	Sets up the evaluation; coordinates with the client, making sure the client's needs are met; establishes the evaluation contract; forms the evaluation team; sees that the final report is produced and delivered.
Evaluation Leader	Runs the evaluation; facilitates elicitation of scenarios; administers the scenario prioritization process; facilitates the evaluation of scenarios against the architecture.
Scenario Scribe	Writes scenarios in a sharable, public form during scenario elicitation; captures the agreed-on wording of each scenario, halting discussion until the exact wording is captured.
E-Scribe	Captures the proceedings in electronic form: raw scenarios, issue(s) that motivate each scenario (often lost in the wording of the scenario itself), and the results of each scenario's analysis; also generates a list of adopted scenarios for distribution to all participants.
Questioner	Asks probing quality attribute–based questions.

Outputs of the ATAM

1. *A concise presentation of the architecture.* One requirement of the ATAM is that the architecture be presented in one hour or less, which leads to an architectural presentation that is both concise and, usually, understandable.
2. *Articulation of the business goals.* Frequently, the business goals presented in the ATAM exercise are being seen by some of the assembled participants for the first time and these are captured in the outputs. This description of the business goals survives the evaluation and becomes part of the project's legacy.
3. *Prioritized quality attribute requirements expressed as quality attribute scenarios.* These quality attribute scenarios take the form described in Chapter 3. The ATAM uses prioritized quality attribute scenarios as the basis for evaluating the architecture. Those scenarios may already exist (perhaps as a result of a prior requirements-capture exercise or ADD activity), but if not, they are generated by the participants as part of the ATAM exercise.
4. *A set of risks and non-risks.* An architectural risk is a decision that may lead to undesirable consequences in light of stated quality attribute requirements. Similarly, an architectural non-risk is a decision that, upon analysis, is deemed safe. The identified

risks form the basis for an architectural risk mitigation plan. These risks are the primary output of an ATAM exercise.

5. *A set of risk themes.* When the analysis is complete, the evaluation team examines the full set of discovered risks to look for overarching themes that identify systemic weaknesses in the architecture or even in the architecture process and team. If left untreated, these risk themes will threaten the project's business goals.
6. *Mapping of architectural decisions to quality requirements.* Architectural decisions can be interpreted in terms of the drivers that they support or hinder. For each quality attribute scenario examined during an ATAM exercise, those architectural decisions that help to achieve it are determined and captured. They can serve as a statement of the rationales for those decisions.
7. *A set of identified sensitivity points and tradeoff points.* Sensitivity points are architectural decisions that have a marked effect on a quality attribute response. Tradeoffs occur when two or more quality attribute responses are sensitive to the same architectural decision, but one of them improves while the other degrades—hence the tradeoff.

The outputs of the ATAM exercise can be used to build a final report that recaps the method, summarizes the proceedings, captures the scenarios and their analysis, and catalogs the findings.

An ATAM-based evaluation also produces intangible results that should not be ignored. These include a sense of community on the part of the stakeholders, open communication channels between the architect and the stakeholders, and a better overall understanding among all participants of the architecture and its strengths and weaknesses. While these results are hard to measure, they are no less important than the others.

Phases of the ATAM

Activities in an ATAM-based evaluation are spread out over four phases:

- In phase 0, “Partnership and Preparation,” the evaluation team leadership and the key project decision makers work out the details of the exercise. The project representatives brief the evaluators about the project so that the evaluation team can be supplemented by people who possess the appropriate expertise. Together, the two groups agree on logistics, such as the time when the evaluation will take place and technology used to support the meetings. They also agree on a preliminary list of stakeholders (by name, not just role), and negotiate when the final report will be delivered and to whom. They deal with formalities such as a statement of work or nondisclosure agreements. The evaluation team examines the architecture documentation to gain an understanding of the architecture and the major design approaches that it comprises. Finally, the evaluation team leader explains what information the manager and architect will be expected to show during phase 1, and helps them construct their presentations, if necessary.
- During Phases 1 and 2, collectively known as “Evaluation,” everyone gets down to the business of analysis. By now, the evaluation team will have studied the architecture documentation and will have a good idea of what the system is about, the major architectural

approaches taken, and the quality attributes that are of paramount importance. During phase 1, the evaluation team meets with the project decision makers to begin information gathering and analysis. In phase 2, the architecture’s stakeholders add their input to the proceedings and analysis continues.

- In Phase 3, “Follow-up,” the evaluation team produces and delivers its final report. This report—which may be a formal document or simply a set of slides—is first circulated to key stakeholders to ensure that it contains no errors of understanding. After this review is complete, it is delivered to the client.

Table 21.2 shows the four phases of the ATAM, who participates in each phase, and the typical cumulative time spent on the activity—possibly in several segments.

TABLE 21.2 ATAM Phases and Their Characteristics

Phase	Activity	Participants	Typical Cumulative Time
0	Partnership and preparation	Evaluation team leadership and key project decision makers	Proceeds informally as required, perhaps over a few weeks
1	Evaluation	Evaluation team and project decision makers	1–2 days
2	Evaluation (continued)	Evaluation team, project decision makers, and stakeholders	2 days
3	Follow-up	Evaluation team and evaluation client	1 week

Source: Adapted from [Clements 01b].

Steps of the Evaluation Phases

The ATAM analysis phases (phases 1 and 2) consist of nine steps. Steps 1–6 are carried out in phase 1 with the evaluation team and the project’s decision makers—typically, the architecture team, project manager, and client. In phase 2, with all stakeholders involved, steps 1–6 are summarized and steps 7–9 are carried out.

Step 1: Present the ATAM

The first step calls for the evaluation leader to present the ATAM to the assembled project representatives. This time is used to explain the process that everyone will be following, to answer questions, and to set the context and expectations for the remainder of the activities. Using a standard presentation, the leader describes the ATAM steps in brief and the outputs of the evaluation.

Step 2: Present the Business Goals

Everyone involved in the evaluation—the project representatives as well as the evaluation team members—needs to understand the context for the system and the primary business goals motivating its development. In this step, a project decision maker (ideally the project manager

or customer representative) presents a system overview from a business perspective. This presentation should describe the following aspects of the project:

- The system's most important functions
- Any relevant technical, managerial, economic, or political constraints
- The business goals and context as they relate to the project
- The major stakeholders
- The architectural drivers (emphasizing architecturally significant requirements)

Step 3: Present the Architecture

The lead architect (or architecture team) makes a presentation describing the architecture at an appropriate level of detail. The “appropriate level” depends on several factors: how much of the architecture has been designed and documented, how much time is available, and the nature of the behavioral and quality requirements.

In this presentation, the architect covers technical constraints such as the operating system, platforms prescribed for use, and other systems with which this system must interact. Most importantly, the architect describes the architectural approaches (or patterns, or tactics, if the architect is fluent in that vocabulary) used to meet the requirements.

We expect architectural views, as introduced in Chapter 1 and described in detail in Chapter 22, to be the primary vehicle by which the architect conveys the architecture. Context diagrams, component-and-connector views, module decomposition or layered views, and the deployment view are useful in almost every evaluation, and the architect should be prepared to show them. Other views can be presented if they contain information relevant to the architecture at hand, especially information relevant to satisfying important quality attribute requirements.

Step 4: Identify the Architectural Approaches

The ATAM focuses on analyzing an architecture by understanding its architectural approaches. Architectural patterns and tactics are useful for (among other reasons) the known ways in which each one affects particular quality attributes. For example, a layered pattern tends to bring portability and maintainability to a system, possibly at the expense of performance. A publish-subscribe pattern is scalable in the number of producers and consumers of data, whereas the active redundancy pattern promotes high availability.

Step 5: Generate a Quality Attribute Utility Tree

The quality attribute goals are articulated in detail via a quality attribute utility tree, which we introduced in Section 19.4. Utility trees serve to make the requirements concrete by defining precisely the relevant quality attribute requirements that the architects were working to provide.

The important quality attribute goals for the architecture under consideration were named or implied in step 2, when the business goals were presented, but not with a degree of specificity that would permit analysis. Broad goals such as “modifiability” or “high throughput” or “ability to be ported to a number of platforms” establish context and direction, and provide a

backdrop against which subsequent information is presented. However, they are not specific enough to let us tell if the architecture suffices to achieve those aims. Modifiable in what way? Throughput that is how high? Ported to what platforms and in how much time? The answers to these kinds of questions are expressed as quality attribute scenarios representing architecturally significant requirements.

Recall that the utility tree is constructed by the architect and the project decision makers. Together, they determine the importance of each scenario: The architect rates the technical difficulty or risk of the scenario (on a H, M, L scale), and the project decision makers rate its business importance.

Step 6: Analyze the Architectural Approaches

The evaluation team examines the highest-ranked scenarios (as identified in the utility tree) one at a time; the architect is asked to explain how the architecture supports each one. Evaluation team members—especially the questioners—probe for the architectural approaches that the architect used to carry out the scenario. Along the way, the evaluation team documents the relevant architectural decisions and identifies and catalogs their risks, non-risks, and tradeoffs. For well-known approaches, the evaluation team asks how the architect overcame known weaknesses in the approach or how the architect gained assurance that the approach sufficed. The goal is for the evaluation team to be convinced that the instantiation of the approach is appropriate for meeting the attribute-specific requirements for which it is intended.

Scenario walkthrough leads to a discussion of possible risks and non-risks. For example:

- The frequency of heartbeats affects the time in which the system can detect a failed component. Some assignments will result in unacceptable values of this response; these are risks.
- The frequency of heartbeats determines the time for detection of a fault.
- Higher frequency leads to improved availability but also consumes more processing time and communication bandwidth (potentially leading to reduced performance). This is a tradeoff.

These issues, in turn, may catalyze a deeper analysis, depending on how the architect responds. For example, if the architect cannot characterize the number of clients and cannot say how load balancing will be achieved by allocating processes to hardware, there is little point in proceeding to any performance analysis. If such questions can be answered, the evaluation team can perform at least a rudimentary, or back-of-the-envelope, analysis to determine if these architectural decisions are problematic vis-à-vis the quality attribute requirements they are meant to address.

The analysis during step 6 is not meant to be comprehensive. The key is to elicit sufficient architectural information to establish some link between the architectural decisions that have been made and the quality attribute requirements that need to be satisfied.

Figure 21.1 shows a template for capturing the analysis of an architectural approach for a scenario. As shown in the figure, based on the results of this step, the evaluation team can identify and record a set of risks and non-risks, sensitivity points, and tradeoffs.

Scenario #: A12		Scenario: Detect and recover from HW failure of main switch.		
Attribute(s)	Availability			
Environment	Normal operations			
Stimulus	One of the CPUs fails			
Response	0.999999 availability of switch			
Architectural decisions	Sensitivity	Tradeoff	Risk	Nonrisk
Backup CPU(s)	S2		R8	
No backup data channel	S3	T3	R9	
Watchdog	S4			N12
Heartbeat	S5			N13
Failover routing	S6			N14
Reasoning	Ensures no common mode failure by using different hardware and operating system (see Risk 8) Worst-case rollover is accomplished in 4 seconds, as computing state takes that long at worst Guaranteed to detect failure within 2 seconds based on rates of heartbeat and watchdog Watchdog is simple and has proved reliable Availability requirement might be at risk due to lack of backup data channel . . . (see Risk 9)			
Architecture diagram	<pre>graph LR; In(()) --> P[Primary CPU OS1]; In --> B[Backup CPU with Watchdog OS2]; P --> S[Switch CPU OS1]; B --> S; P -.-> Heartbeat 1 sec B; S --> Out(())</pre>			

FIGURE 21.1 Example of architecture approach analysis (adapted from [Clements 01b])

At the end of step 6, the evaluation team should have a clear picture of the most important aspects of the entire architecture, the rationale for key design decisions, and a list of risks, non-risks, sensitivity points, and tradeoff points.

At this point, phase 1 is concluded.

Hiatus and Start of Phase 2

The evaluation team summarizes what it has learned and interacts informally with the architect during a hiatus of a week or so. More scenarios might be analyzed during this period, if desired, or answers to questions posed in phase 1 may be clarified.

Attendees at the phase 2 meeting include an expanded list of participants, with additional stakeholders joining the discussion. To use an analogy from programming: Phase 1 is akin to when you test your own program, using your own criteria. Phase 2 is when you give your program to an independent quality assurance group, who will likely subject your program to a wider variety of tests and environments.

In phase 2, step 1 is repeated so that the stakeholders understand the method and the roles they are to play. Then the evaluation leader recaps the results of steps 2–6, and shares the current list of risks, non-risks, sensitivity points, and tradeoffs. After bringing the stakeholders up to speed with the evaluation results so far, the remaining three steps can be carried out.

Step 7: Brainstorm and Prioritize Scenarios

The evaluation team asks the stakeholders to brainstorm quality attribute scenarios that are operationally meaningful with respect to the stakeholders' individual roles. A maintainer will likely propose a modifiability scenario, while a user will probably come up with a scenario that expresses ease of operation, and a quality assurance person will propose a scenario about testing the system or being able to replicate the state of the system leading up to a fault.

While utility tree generation (step 5) is used primarily to understand how the architect perceived and handled quality attribute architectural drivers, the purpose of scenario brainstorming is to take the pulse of the larger stakeholder community: to understand what system success means for them. Scenario brainstorming works well in larger groups, creating an atmosphere in which the ideas and thoughts of one person stimulate others' ideas.

Once the scenarios have been collected, they must be prioritized, for the same reasons that the scenarios in the utility tree needed to be prioritized: The evaluation team needs to know where to devote its limited analysis time. First, stakeholders are asked to merge scenarios they feel represent the same behavior or quality concern. Next, they vote for those scenarios they feel are most important. Each stakeholder is allocated a number of votes equal to 30 percent of the number of scenarios,¹ rounded up. Thus, if 40 scenarios were collected, each stakeholder would be given 12 votes. These votes can be allocated in any way that the stakeholder sees fit: all 12 votes for 1 scenario, 1 vote for each of 12 distinct scenarios, or anything in between.

The list of prioritized scenarios is compared with those from the utility tree exercise. If they agree, it indicates good alignment between what the architect had in mind and what the stakeholders actually wanted. If additional driving scenarios are discovered—and they usually are—this may itself be a risk, if the discrepancy is large. Such discoveries indicate some level of disagreement about the system's important goals between the stakeholders and the architect.

1. This is a common facilitated brainstorming technique.

Step 8: Analyze the Architectural Approaches

After the scenarios have been collected and prioritized in step 7, the evaluation team guides the architect in the process of analyzing the highest-ranked scenarios. The architect explains how architectural decisions contribute to realizing each scenario. Ideally, this activity will be dominated by the architect's explanation of scenarios in terms of previously discussed architectural approaches.

In this step the evaluation team performs the same activities as in step 6, using the highest-ranked, newly generated scenarios. Typically, this step might cover the top five to ten scenarios, as time permits.

Step 9: Present the Results

In step 9, the evaluation team convenes and groups risks into risk themes, based on some common underlying concern or systemic deficiency. For example, a group of risks about inadequate or out-of-date documentation might be grouped into a risk theme stating that documentation is given insufficient consideration. A group of risks about the system's inability to function in the face of various hardware and/or software failures might lead to a risk theme about insufficient attention to backup capability or providing high availability.

For each risk theme, the evaluation team identifies which of the business goals listed in step 2 are affected. Identifying risk themes and then relating them to specific drivers brings the evaluation full circle by relating the final results to the initial presentation, thereby providing a satisfying closure to the exercise. Equally important, it elevates the risks that were uncovered to the attention of management. What might otherwise have seemed to a manager like an esoteric technical issue is now identified unambiguously as a threat to something the manager is on record as caring about.

The collected information from the evaluation is summarized and presented to stakeholders. The following outputs are presented:

- The architectural approaches documented
- The set of scenarios and their prioritization from the brainstorming
- The utility tree
- The risks and non-risks discovered
- The sensitivity points and tradeoffs found
- Risk themes and the business goals threatened by each one

Going Off Script

Years of experience have taught us that no architecture evaluation exercise ever goes completely by the book. And yet for all the ways that an exercise might go terribly wrong, for all the details that can be overlooked, for all the fragile egos that can be bruised, and for all the high stakes that are on the table, we have never had an architecture evaluation exercise spiral out of control. Every single one has been a success, as measured by the feedback we gather from clients.

While they all turned out successfully, there have been a few memorable cliffhangers.

More than once, we began an architecture evaluation, only to discover that the development organization had no architecture to be evaluated. Sometimes there was a stack of class diagrams or vague text descriptions masquerading as an architecture. Once we were promised that the architecture would be ready by the time the exercise began, but in spite of good intentions, it wasn't. (We weren't always so prudent about pre-exercise preparation and qualification. Our current diligence is a result of experiences like these.) But it was okay. In cases like these, the evaluation's main results included the articulated set of quality attributes, a "whiteboard" architecture sketched during the exercise, plus a set of documentation obligations for the architect. In all cases, the client felt that the detailed scenarios, the analysis we were able to perform on the elicited architecture, and the recognition of what needed to be done more than justified the exercise.

A couple of times we began an evaluation, only to lose the architect in the middle of the exercise. In one case, the architect resigned between preparation and execution of the evaluation. This organization was in turmoil, and the architect simply got a better offer in a calmer environment elsewhere. Usually, we don't proceed without the architect, but it was okay, because the architect's apprentice stepped in. A little additional prework to prepare him, and we were all set. The evaluation went off as planned, and the preparation that the apprentice did for the exercise helped mightily to prepare him to step into the architect's shoes.

Once we discovered halfway through an ATAM exercise that the architecture we had prepared to evaluate was being jettisoned in favor of a new one that no one had bothered to mention. During step 6 of phase 1, the architect responded to a problem raised by a scenario by casually mentioning that "the new architecture" would not suffer from that deficiency. Everyone in the room, stakeholders and evaluators alike, looked at each other in the puzzled silence that followed. "What new architecture?" I asked blankly, and out it came. The developing organization (a contractor for the U.S. military, which had commissioned the evaluation) had prepared a new architecture for the system to handle the more stringent requirements they knew were coming in the future. We called a timeout, conferred with the architect and the client, and decided to continue the exercise using the new architecture as the subject instead of the old. We backed up to step 3 (the architecture presentation), but everything else on the table—business goals, utility tree, scenarios—remained completely valid. The evaluation proceeded as before, and at the conclusion of the exercise, our military client was extremely pleased at the knowledge gained.

In perhaps the most bizarre evaluation in our experience, we lost the architect midway through phase 2. The client for this exercise was the project manager in an organization undergoing a massive restructuring. The manager was a pleasant gentleman with a quick sense of humor, but there was an undercurrent that said he was not to be crossed. The architect was being reassigned to a different part of the organization in the near future; this was tantamount to being fired from the project, and the manager said he wanted to establish the quality of the architecture before his architect's awkward departure. (We didn't find any of this out until after the evaluation.) When we set up the ATAM exercise, the manager suggested that the junior designers attend. "They might learn something," he said. We agreed. As the exercise began, our schedule (which was very tight to begin with) kept being disrupted. The manager wanted us to meet with his company's executives. Then he wanted us to have a long lunch with someone who he said could give us more architectural insights. It turned out that the executives were busy at the time of our scheduled meeting. So the manager asked if we could come back and meet with them later on.

By now, phase 2 was thrown off schedule to such an extent that the architect, to our horror, had to leave to fly back to his home in a distant city. He was none too happy that his architecture was going to be evaluated without him. The junior designers, he said, would never be able to answer our questions. Before his departure, our team huddled. The exercise seemed to be teetering on the brink of disaster. We had an unhappy departing architect, a blown schedule, and questionable expertise available. We decided to split our evaluation team. One half of the team would continue with phase 2 using the junior designers as our information resource. The second half of the team would continue with phase 2 by telephone the next day with the architect. Somehow we would make the best of a bad situation.

Surprisingly, the project manager seemed completely unperturbed by the turn of events. "It will work out, I'm sure," he said pleasantly, and then retreated to confer with various vice presidents about the reorganization.

I led the team interviewing the junior designers. We had never gotten a completely satisfactory architecture presentation from the architect. Discrepancies in the documentation were met with a breezy "Oh, well, that's not how it really works." So I decided to start over with ATAM step 3. We asked the half dozen or so designers what their view of the architecture was. "Could you draw it?" I asked them. They looked at each other nervously, but one said, "I think I can draw part of it." He took to the whiteboard and drew a very reasonable component-and-connector view. Someone else volunteered to draw a process view. A third person drew the architecture for an important offline part of the system. Others jumped in to assist.

As we looked around the room, everyone was busy transcribing the whiteboard pictures. None of the pictures corresponded to anything we had seen in the documentation so far. "Are these diagrams documented anywhere?" I asked. One of the designers looked up from his busy scribbling for a moment to grin. "They are now," he said.

As we proceeded to step 8, analyzing the architecture using the scenarios previously captured, the designers did an astonishingly good job of working together to answer our questions. Nobody knew everything, but everybody knew something. Together in a half day, they produced a clear and consistent picture of the whole architecture that was much more coherent and understandable than anything the architect had been willing to produce in two whole days of pre-exercise discussion. And by the end of phase 2, the design team was transformed. This erstwhile group of information-starved individuals with limited compartmentalized knowledge became a true architecture team. The members drew out and recognized each other's expertise. This expertise was revealed and validated in front of everyone—and most important, in front of their project manager, who had slipped back into the room to observe. There was a look of supreme satisfaction on his face. It began to dawn on me that—you guessed it—it was okay.

It turned out that this project manager knew how to manipulate events and people in ways that would have impressed Machiavelli. The architect's departure was not because of the reorganization, but merely coincident with it. The project manager had orchestrated it. The architect had, the manager felt, become too autocratic and dictatorial, and the manager wanted the junior design staff to be given the opportunity to mature and contribute. The architect's mid-exercise departure was exactly what the project manager had wanted. And the design team's emergence under fire had been the primary purpose of the evaluation exercise all along. Although we found several important issues related to the architecture, the project manager knew about every one

of them before we ever arrived. In fact, he made sure we uncovered some of them by making a few discreet remarks during breaks or after a day's session.

Was this exercise a success? The client could not have been more pleased. His instincts about the architecture's strengths and weaknesses were confirmed. We were instrumental in helping his design team, which would guide the system through the stormy seas of the company's reorganization, come together as an effective and cohesive unit at exactly the right time. And the client was so pleased with our final report that he made sure the company's board of directors saw it.

These cliffhangers certainly stand out in our memory. There was no architecture documented. But it was okay. It wasn't the right architecture. But it was okay. There was no architect. But it was okay. The client really wanted to effect a team reorganization. In every instance, we reacted as reasonably as we could, and each time it was okay.

Why? Why, time after time, does it turn out okay? I think there are three reasons.

First, the people who commission the architecture evaluation really want it to succeed. The architect, developers, and stakeholders assembled at the client's behest also want it to succeed. As a group, they help keep the exercise marching toward the goal of architectural insight.

Second, we are always honest. If we feel that the exercise is derailing, we call a timeout and confer among ourselves, and usually confer with the client. While a small amount of bravado can come in handy during an exercise, we never, ever try to bluff our way through an evaluation. Participants can detect that false note instinctively, and the evaluation team must never lose the respect of the other participants.

Third, the methods are constructed to establish and maintain a steady consensus throughout the exercise. There are no surprises at the end. The participants lay down the ground rules for what constitutes a suitable architecture, and they contribute to the risks uncovered at every step of the way.

So: Do the best job you can. Be honest. Trust the methods. Trust in the goodwill and good intentions of the people you have assembled. And it will be okay.

—PCC (Adapted from [Clements 01b])

21.6 Lightweight Architecture Evaluation

The Lightweight Architecture Evaluation (LAE) method is intended to be used in a project-internal context where the reviewing is carried out by peers on a regular basis. It uses the same concepts as the ATAM and is meant to be performed regularly. An LAE session may be convened to focus on what has changed since the prior review—in the architecture or in the architecture drivers—or to examine a previously unexamined portion of the architecture. Because of this limited scope, many of the ATAM's steps can be omitted or shortened.

The duration of an LAE exercise depends on the number of quality attribute scenarios generated and examined, which is in turn based on the scope of the review. The number of scenarios examined depends on the importance of the system being reviewed. Thus an LAE exercise can be as short as a couple of hours or as long as a full day. It is carried out entirely by members internal to the organization.